Created by Adm Malik

Solved by Zennia Khan

Bt102

Microbial identification

Microbial growth control

Karyogamy, plasmogamy, meiosis

Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-).

Karyogamy: The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

Meiosis: The diploid nucleus gives rise to haploid nuclei (sexual spores)

Transduction, transformation, conjugation

Conjugation:-

Mediated by a plasmid
• Plasmid is an extra chromosomal circular DNA fragment. Replicates independently of chromosome. Genes of plasmids not important for bacterial growth.

• Requires direct cell to cell contact. Opposite mating types cells. Plasmids in G –ive cells have genes that code for sex pili. G +ve secrete sticky surface molecules

These plasmids are transmissible from one bacterium to another during conjugation.

Conjugation is mediated by sex pili and plasmids or F factors can be transferred.

Conjugation can be used for gene mapping on the chromosomes.
• conjugation_mapping.html

• High frequency of recombination is also a kind of conjugation.

Transformation can occur in vitro

Transformation is uptake of naked DNA by microbes

If you need any other queries If you need any other queries, then freely to ask us

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Difference b/w viable and direct count

Viable counting method
– Spread plate
– Pour plate

• Pour plate and spread methods

counting method

– Spread plate with serial dilutions

Direct counts
• Counting chamber
• Coulter Counter
• Flow cytometer

• Membrane filter technique for water samples

Economical use of fungi

Immunity, adaptive and innate immunity

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Innate: Present at birth

▪ **Adaptive:** Acquired by experience

Innate Immunity
▪ Present at birth

▪ Rapid to respond

▪ Nonspecific

▪ Do not increase in response with repeated exposures to the same agent

▪ Have no memory

Innate Immunity
▪ **First line of defense:** Physical barriers (skin, mucus membrane)

▪ **Second line of defense:** Phagocytic cells and their secretions

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Adaptive Immunity

- Specific
- Have memory
- Increase with repeated exposure
- Slower to respond
- **Humoral immune response**
- Cellular immune response

Bt102

What are buffers?¨

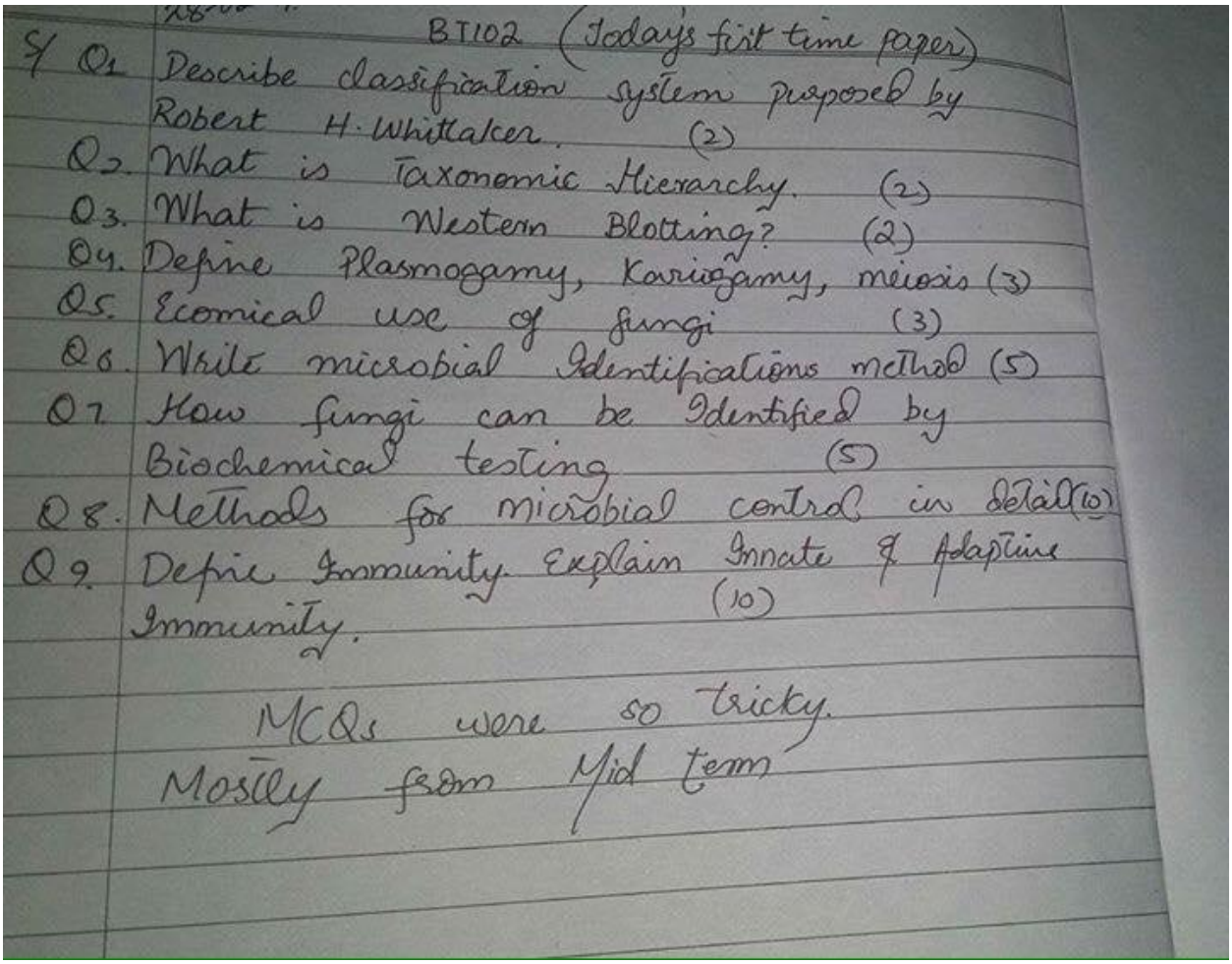
Buffers are used to maintain pH in medium.
Buffers are used to maintain desired pH

- Peptones and amino acids
- PBS

How infectious disease can be transmitted explain it briefly transmission occur to contact, vehicle, vector?

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Bt102
Characters tics of aerophiles (2)
Difference between vertical and horizontal gene transfer (3)

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